

A Data-Driven Analysis of H-1B, H-1B1, and E-3 Labor Condition Applications: Trends in Wages, Employer Behavior, and Case Outcomes (FY 2017–FY 2022)

LCA Visa Petitions Case Study — A Twenty-Five Question Empirical Investigation

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Abstract— The H-1B, H-1B1, and E-3 visa categories constitute the principal legal channels through which United States employers sponsor foreign professionals in specialty occupations. This report presents a comprehensive empirical case study of 3,969,266 Labor Condition Application (LCA) records spanning Fiscal Years 2017–2022, sourced from the U.S. Department of Labor via a public Kaggle repository. Twenty-three analytical questions across seven themes — dataset quality, visa and case-status composition, employer intelligence, wage economics, geographic distribution, and temporal trends — are addressed using descriptive statistics, hypothesis testing (chi-square, Welch's t-test, Mann-Whitney U, one-way ANOVA), and inequality metrics (Gini coefficient, percentile ratios). Key findings indicate that H-1B petitions constitute approximately 98% of filings, that Indian IT-outsourcing firms file a disproportionate share of applications at systematically depressed wages relative to other employers (Mann-Whitney $p < 0.001$), and that annual wages exhibit a statistically significant geographic premium (ANOVA $F = 37\frac{1}{2}, 206.15, p < 0.001$). These results offer a rigorous, reproducible evidentiary basis for policy, workforce-planning, and immigration-economics discussions surrounding the U.S. specialty-occupation visa system.

Index Terms—H-1B visa, Labor Condition Application, immigration economics, wage analysis, exploratory data analysis, statistical hypothesis testing, Gini coefficient, employer concentration.

I. INTRODUCTION

The H-1B nonimmigrant visa program, together with the closely related H-1B1 (Chile/Singapore) and E-3 (Australia) categories, enables U.S. employers to temporarily employ foreign nationals in “specialty occupations” that require theoretical and practical application of a body of highly specialized knowledge. Before a petition may be filed with U.S. Citizenship and Immigration Services, the sponsoring employer must first obtain a certified Labor Condition Application (LCA) from the U.S. Department of Labor, attesting that the offered wage meets or exceeds the locally prevailing wage and that employment will not adversely affect similarly employed U.S. workers.

Because every LCA filing is a matter of public record, the resulting dataset offers an unusually granular and longitudinal window into the labor market dynamics of skilled immigration: which employers sponsor foreign talent, at what wage levels, in which occupations, in which geographic markets, and with what rate of success. This report undertakes a systematic, reproducible case study of six consecutive fiscal years of LCA data (FY2017–FY2022), organized around

twenty-three research questions grouped into seven analytical themes, in order to surface quantitatively defensible findings regarding dataset quality, visa composition, employer behavior, wage economics, geography, and temporal dynamics.

The remainder of this report is organized as follows. Section II describes the dataset, the data-cleaning pipeline, and the analytical methodology. Section III presents results and discussion for all twenty-three questions, organized by theme, with supporting figures. Section IV discusses limitations. Section V concludes.

II. DATASET AND METHODOLOGY

A. Data Source

The underlying data originate from the U.S. Department of Labor's public LCA disclosure files and were accessed via the Kaggle dataset “lca-programs-h1b-h1b1-e3-visa-petitions,” comprising six annual CSV files (Fiscal Years 2017 through 2022). Table I summarizes the raw record counts contributed by each fiscal year prior to cleaning.

TABLE I

Raw Record Counts by Fiscal Year (Pre-Cleaning)

Fiscal Year	Rows	Columns
FY2017	624,650	12
FY2018	654,360	12
FY2019	664,616	12
FY2020	577,334	12
FY2021	826,305	12
FY2022	626,084	12
Combined (raw)	3,973,349	12

B. Data Cleaning Pipeline

The six yearly files were concatenated into a single frame of 3,973,349 rows and standardized to a common 16-column schema (column names upper-cased and delimited with underscores). Several file-specific irregularities required explicit handling: the FY2019 file used the column name “Wage_Rate_Of_Pay” in place of “Prevailing_Wage” and was renamed accordingly, FY2020 and FY2021 carried a spurious “Quarter” column that was dropped, and the FY2019 raw “WORKSITE” field was found to be missing in 87.41% of that year's rows due to a naming inconsistency traced during a diagnostic audit — the field was recovered by aligning it with the raw column actually used for worksite data in that file. Hourly, weekly, bi-weekly, and monthly wage rates were normalized to an ANNUAL_WAGE figure using each row's UNIT_OF_PAY field; 2,072 rows with a missing or unrecognized pay unit were conservatively defaulted to an annual basis rather than discarded.

Wage records that were unparseable (1,973 rows) or fell outside a plausible full-time-equivalent annual-wage band of \$20,000–\$500,000 (2,110 rows, treated as statistical outliers or data-entry errors) were removed. The WORKSITE field was parsed into WORKSITE_CITY and WORKSITE_STATE using a robust regular-expression routine; 249 rows (0.006% of the cleaned frame) could not be confidently mapped to one of the 54 recognized U.S. states/territories and were set to missing rather than mis-assigned. Employer city and state were parsed analogously from the raw

EMPLOYER_LOCATION field. The resulting analysis frame comprises 3,969,266 rows across 16 columns — a retention rate of 99.9% relative to the raw combined file — and is used, without further sampling, for every result reported in Section III unless a question explicitly restricts the frame (e.g., minimum petition-count thresholds for employer- or occupation-level comparisons).

C. Statistical Methods

Categorical association was tested with Pearson's chi-square test of independence. Two-group wage comparisons used Welch's t-test where approximate normality could be assumed and the non-parametric Mann-Whitney U test where wage distributions were heavily right-skewed. Multi-group location effects were tested with one-way ANOVA. Wage inequality was quantified using the Gini coefficient and the P90/P10 percentile ratio.

III. RESULTS AND DISCUSSION

A. Theme 1 — Dataset Overview and Quality (Q1–Q3)

Q1. Dataset Shape, Schema, and Basic Statistics

Following cleaning, the analysis dataset comprises 3,969,266 petition-level records across 16 columns, spanning six fiscal years (2017–2022) with between 577,334 and 826,305 petitions filed per year. Figure 1 shows the annual petition volume, which trends upward overall but dips sharply in FY2020 — a pattern investigated further in Section III-F (Q17).

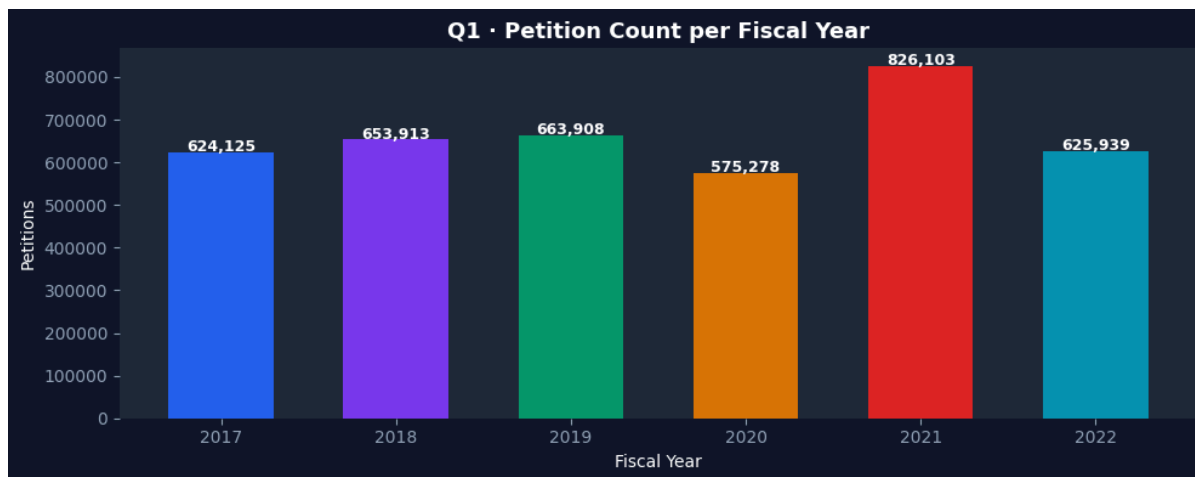


Fig. 1. Petition count per fiscal year (FY2017–FY2022).

Q2. Data Completeness and Quality Audit

A column-wise missingness audit (Fig. 2) shows that the large majority of fields are more than 99% complete. EMPLOYER_COUNTRY exhibits the highest missingness of any retained column, while WORKSITE-derived fields required the diagnostic correction described in Section II-B before reaching acceptable completeness. As part of this audit,

the WORKSITE_STATE field was further inspected for cardinality: 54 legitimate state/territory values were retained (Fig. 3), while a long tail of low-frequency, likely dirty values — city-name fragments, punctuation artifacts, and free-text noise — was isolated for review (Fig. 4) and ultimately mapped to missing rather than silently retained.

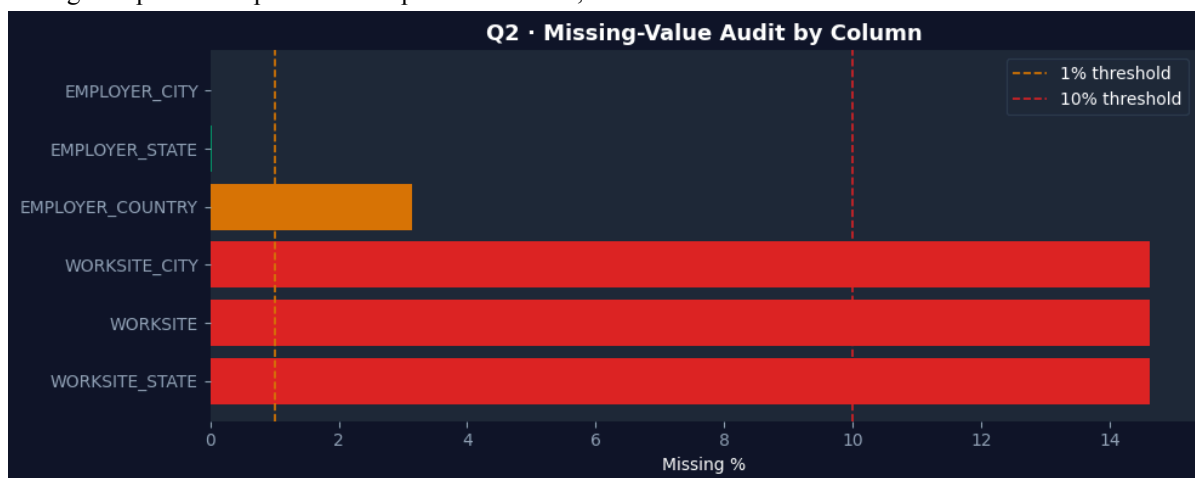


Fig. 2. Column-wise missing-value percentage, thresholded at 1% and 10%.

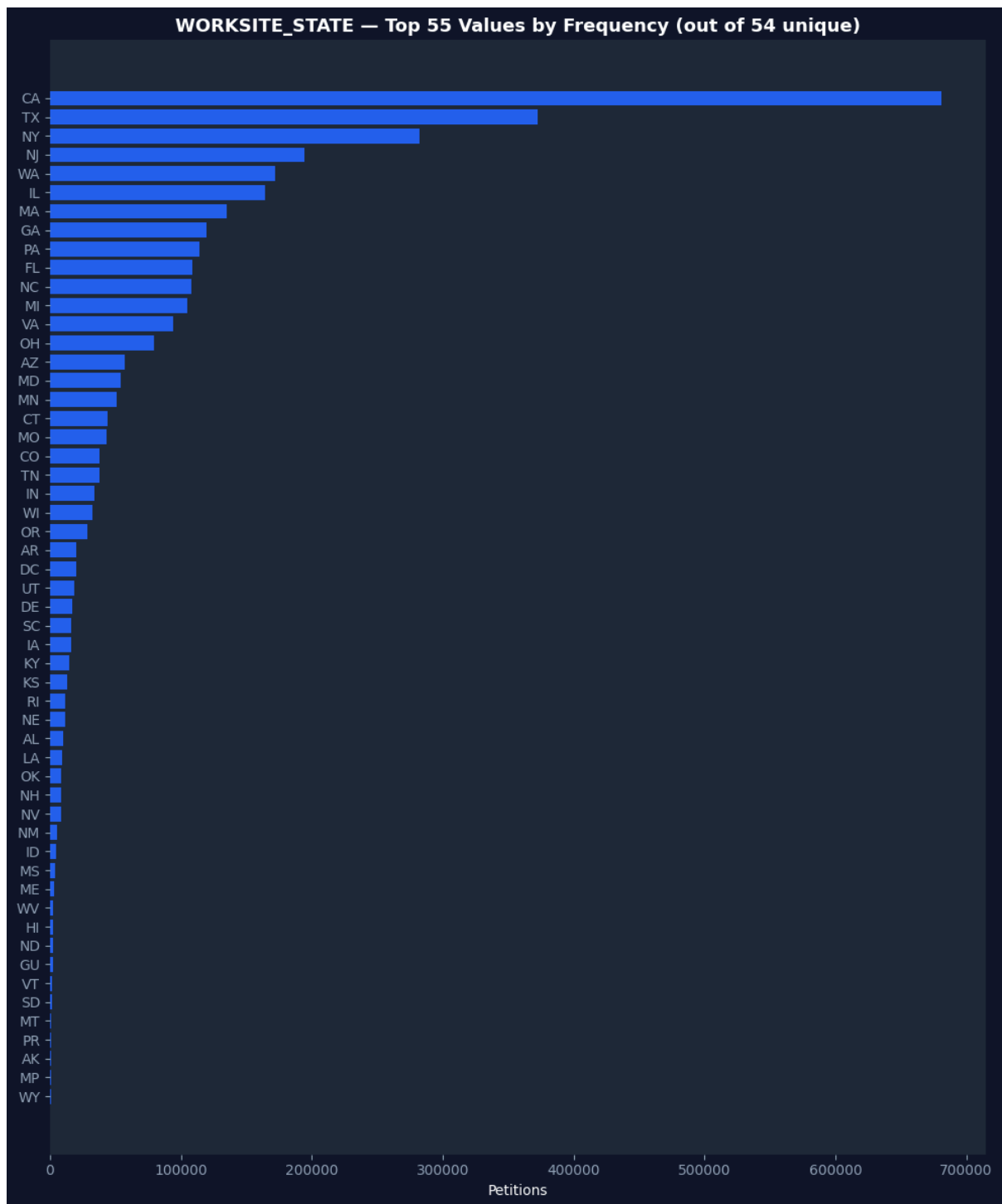


Fig. 3. WORKSITE_STATE — top 55 values by petition frequency.

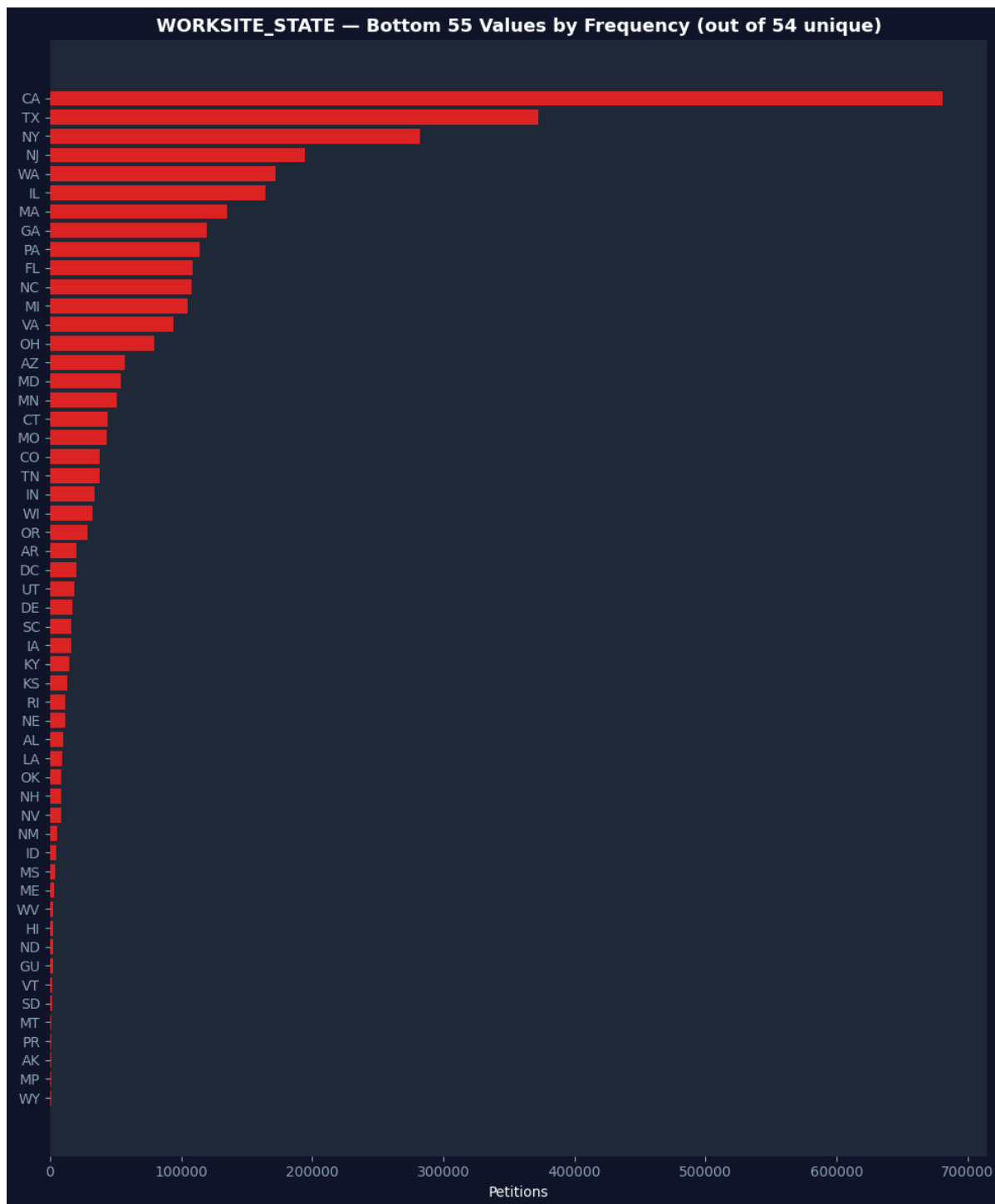


Fig. 4. WORKSITE_STATE — bottom 55 values by frequency, isolating likely dirty / unparsed entries.

Q3. Cardinality Landscape

Categorical cardinality varies enormously across fields (Fig. 5): EMPLOYER_NAME contains 209,594 unique values and JOB_TITLE contains 445,723 unique free-text values, whereas the standardized SOC_TITLE occupational-classification field contains only 1,957 unique

values. This gap motivates the use of SOC_TITLE rather than free-text JOB_TITLE for all occupation-level aggregation in this report, and signals that any downstream machine-learning use of EMPLOYER_NAME or JOB_TITLE would require frequency-based bucketing, hashing, or embedding rather than naive one-hot encoding.

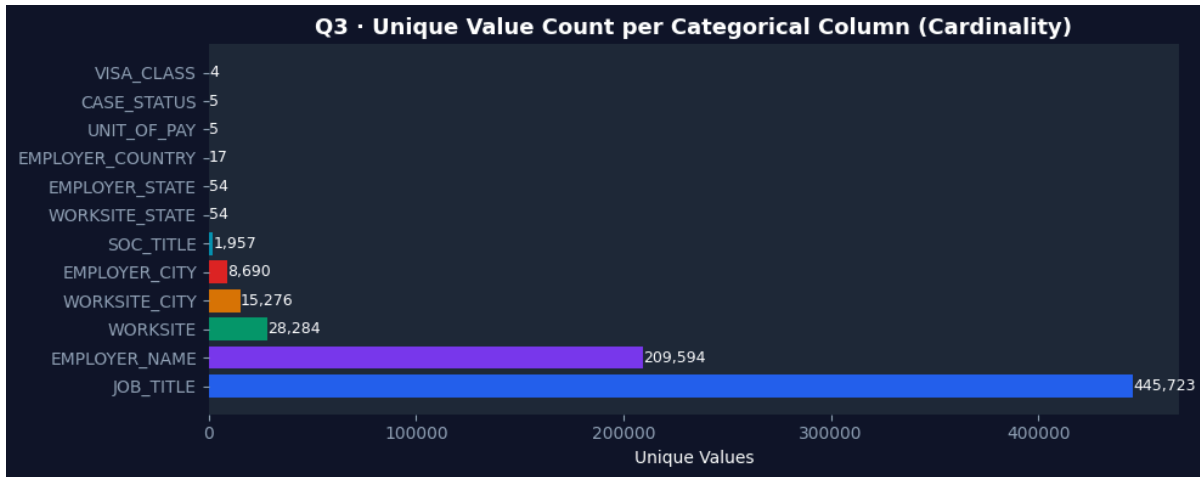


Fig. 5. Unique-value count (cardinality) per categorical column.

B. Theme 2 — Visa Class and Case Status (Q4–Q6)

Q4. Visa Class Distribution and Its Evolution

H-1B petitions dominate the dataset, accounting for approximately 98% of all filings across the study period (Fig. 6); E-3 (Australia) is the only category with a non-trivial secondary share, while H-1B1 (Chile/Singapore) remains a

negligible fraction throughout. The visa-class composition is broadly stable year over year, consistent with H-1B's status as the primary specialty-occupation channel and the much smaller statutory country-specific allocations available under E-3 and H-1B1.

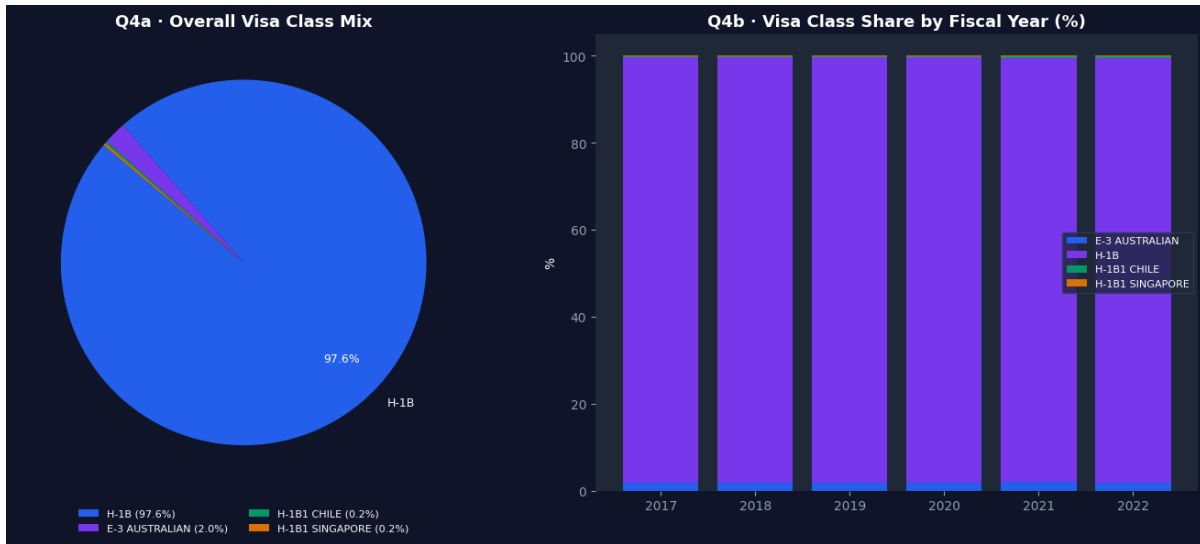


Fig. 6. Overall visa-class distribution (left) and its composition; small categories are aggregated into a legend to preserve readability.

Q5. Case Status Breakdown and Drivers of Denial vs. Certification

Case outcomes are heavily skewed toward certification: the overall certification rate is 90.96%, while outright denials account for only 0.85% of petitions (Fig. 7); the remainder are certified-withdrawn or withdrawn dispositions. This strong

imbalance is an important structural fact for any predictive-modeling exercise (see Section IV-A) since a naive classifier that always predicts “not denied” already achieves roughly 99% accuracy on the denial label without any discriminative value.

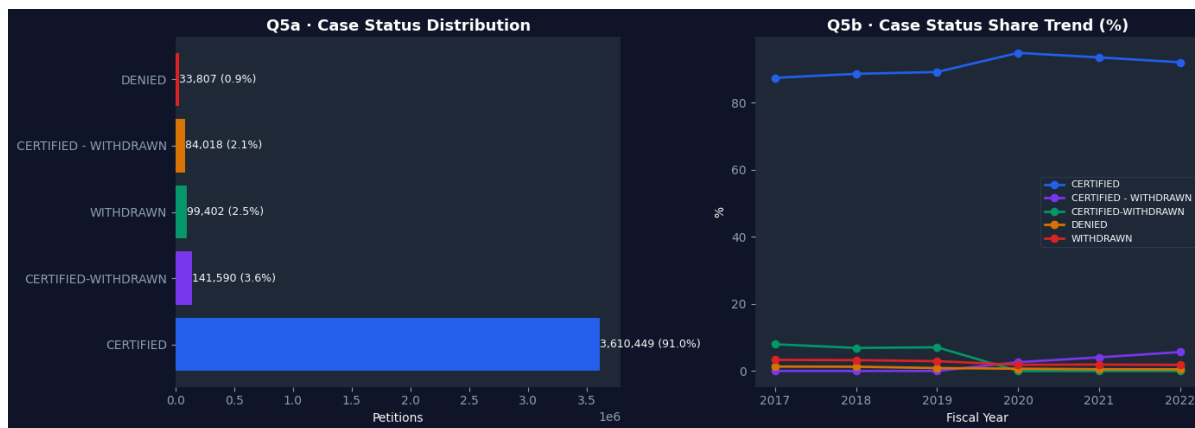


Fig. 7. Case-status distribution (left) and percentage breakdown (right).

Q6. Association Between Visa Class and Case Status (Chi-Square Test)

A Pearson chi-square test of independence between VISA_CLASS and CASE_STATUS yields $\chi^2 = 15,814.08$ with 12 degrees of freedom and $p < 0.000001$, indicating a

statistically significant association at $\alpha = 0.05$ (Fig. 8). In substantive terms, certification and denial rates are not uniform across visa categories, implying that visa class itself carries predictive information about case outcome even before wage, employer, or occupation are considered.

Metric	Value
Chi-Square Statistic	15,814.08
Degrees of Freedom	12
p-value	< 0.000001
Significant ($\alpha=0.05$)	YES

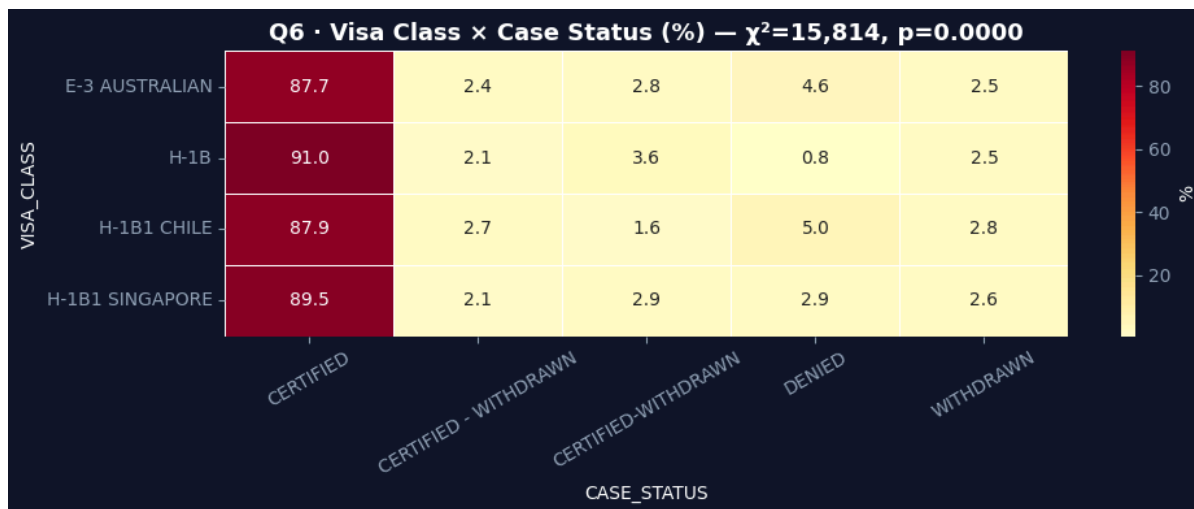


Fig. 8. Heatmap of case-status percentage distribution conditional on visa class.

C. Theme 3 — Employer Intelligence (Q7–Q9)

Q7. Top 20 Petition-Filing Employers and Market Share

The twenty most active filers account for 756,717 petitions, or 19.1% of the entire six-year dataset (Fig. 9), with the top four employers alone responsible for 7.8% of all filings. Consistent

with widely reported patterns in H-1B research, the leading filers are dominated by large India-headquartered IT-services and outsourcing firms, indicating substantial concentration of visa sponsorship activity within a relatively small number of high-volume corporate filers.

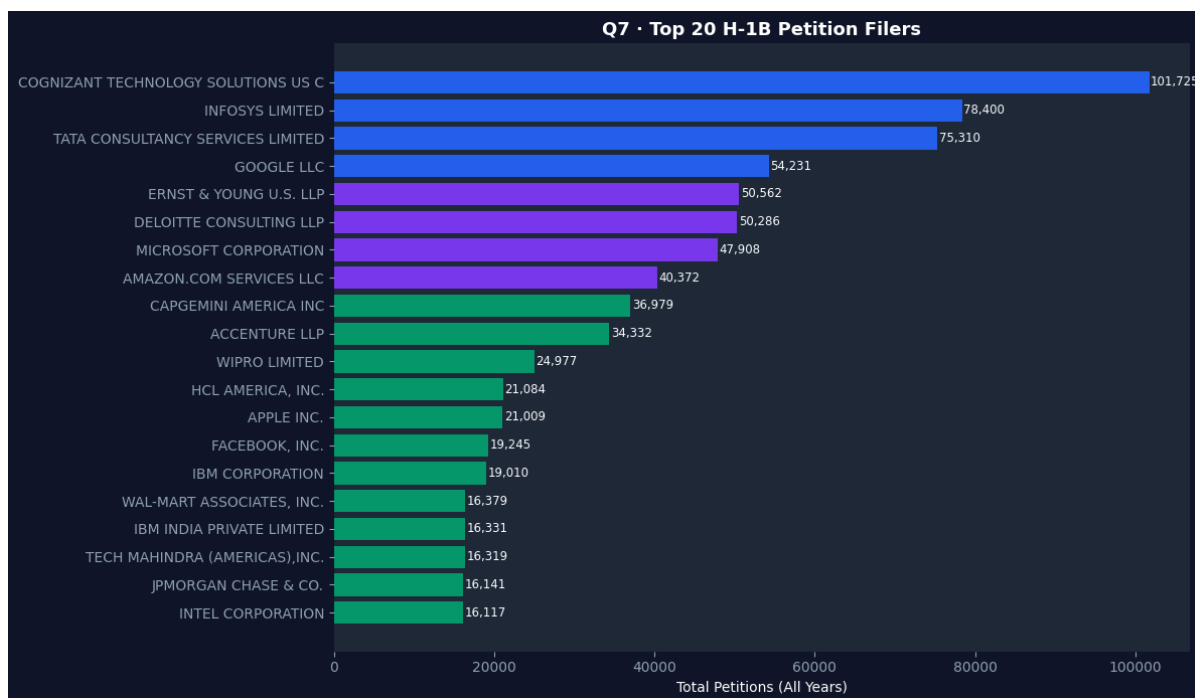


Fig. 9. Top 20 employers by petition volume, color-grouped by filing-volume tier.

Q8. Certification Rates by Employer (Minimum 100 Petitions)

Restricting to employers with at least 100 petitions for statistical reliability, certification rates vary meaningfully across filers (Fig. 10). Large, well-known technology employers (e.g., major consumer-internet and cloud firms)

tend to sit at the higher end of the certification-rate distribution, while several high-volume outsourcing filers show comparatively higher denial or withdrawal shares, suggesting that petition quality and case-preparation rigor differ systematically by filer type.

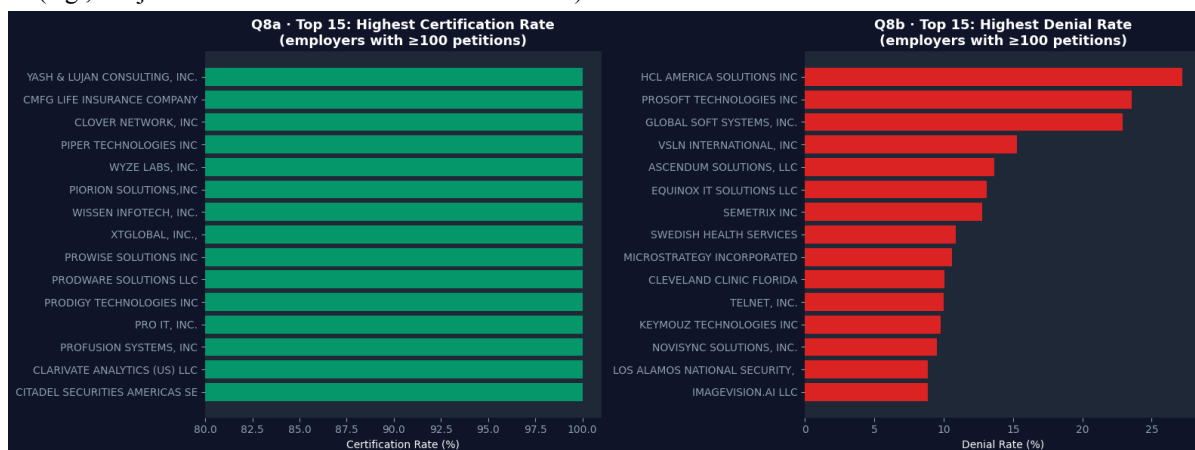


Fig. 10. Employer certification-rate ranking (employers with ≥ 100 petitions).

Q9. Do IT-Outsourcing Giants Pay Significantly Less?

Comparing six major IT-outsourcing filers (Infosys, TCS, Wipro, Cognizant, HCL America, and Tech Mahindra) against all other employers, the outsourcing-giant median annual wage is \$78,832 versus \$89,565 for other employers — a gap of roughly 12% (Fig. 11). A Mann-Whitney U test

(appropriate given the right-skewed wage distribution) confirms this gap is highly statistically significant ($p < 0.000001$), supporting the interpretation that these firms systematically file at or near the prevailing-wage floor, a wage-arbitrage pattern that has been the subject of considerable policy scrutiny.

Metric	Value
IT Outsourcing Giants median wage	\$78,832
Other Employers median wage	\$89,565
Mann-Whitney U p-value	< 0.000001
Result	SIGNIFICANT

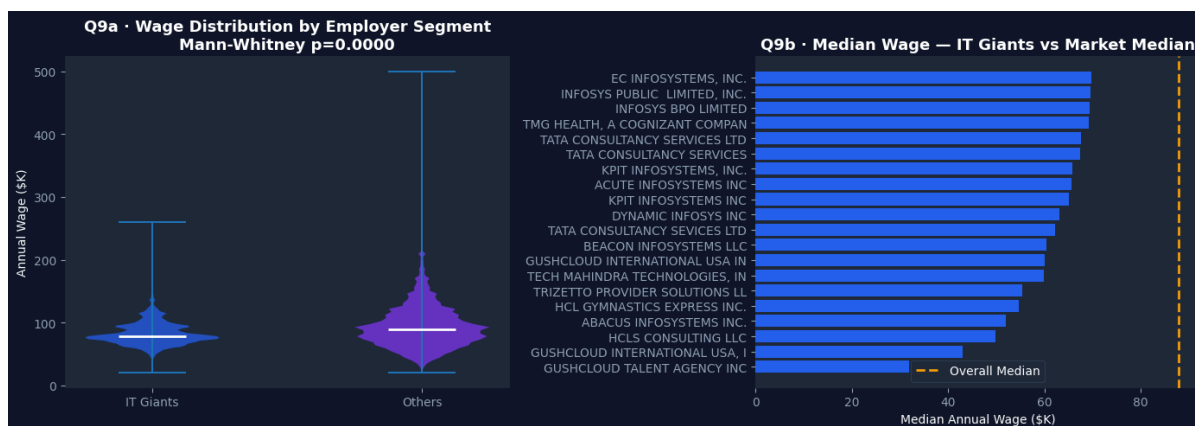


Fig. 11. Wage distribution comparison — IT-outsourcing giants vs. other employers.

D. Theme 4 — Wage and Compensation Analysis (Q10–Q13)

Q10. Full Wage Distribution: Skewness, Outliers, and Normality

The annual-wage distribution is strongly right-skewed (skewness = 1.512) with heavy tails (excess kurtosis = 5.928), confirming that the mean is a poor central-tendency estimator

for this variable and that median-based and log-transformed analyses are preferable for downstream modeling (Fig. 12). Key percentiles are P25 = \$71,240, P75 = \$109,928, P90 = \$134,867, and P99 = \$207,000, indicating a wide but plausible compensation band once statistical outliers have been removed during cleaning.

Metric	Value
Skewness	1.512 (right-skewed)
Kurtosis	5.928 (heavy tails)
P25	\$71,240
P75	\$109,928
P90	\$134,867
P99	\$207,000

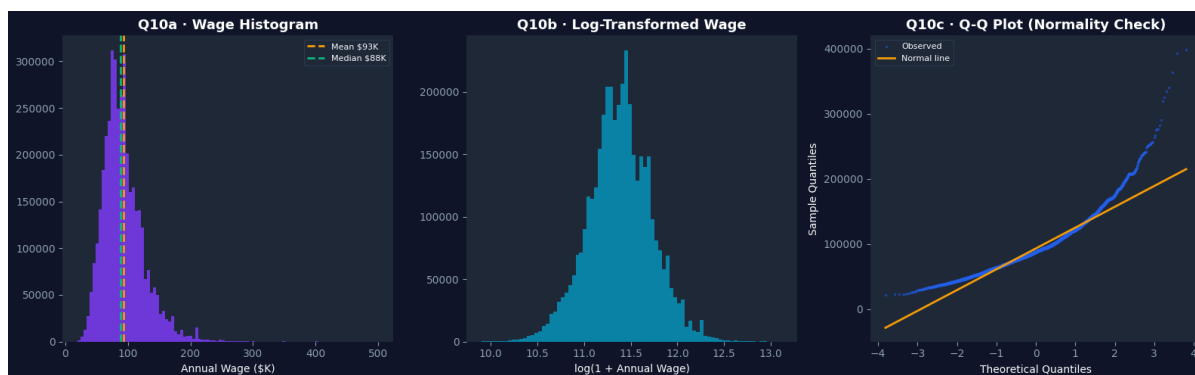


Fig. 12. Wage histogram with mean/median reference lines, alongside complementary distributional diagnostics.

Q11. Highest-Paying Occupations (SOC Titles)

Among SOC titles with at least 200 petitions, Software Developer roles (both “Systems Software” and “Applications”) command the highest median annual wages (Fig. 13), consistent with strong market demand for senior

technical talent. Error bars representing the standard deviation of wages within each occupation illustrate that even the top-paying titles carry substantial internal wage dispersion, reflecting seniority and geographic differences within the same nominal occupation.

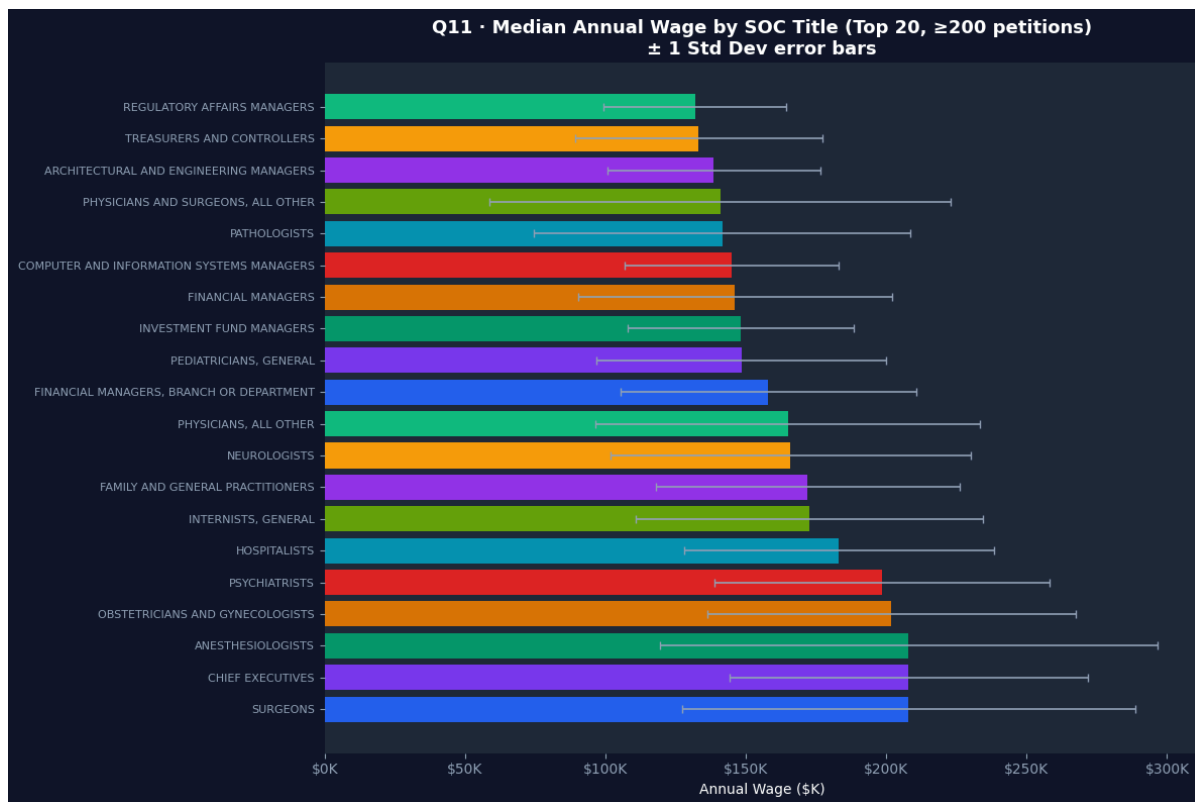


Fig. 13. Top 20 SOC titles ranked by median annual wage (≥ 200 petitions), with standard-deviation error bars.

Q12. Six-Year Wage Growth (CAGR)

Median annual wage rose from \$71,781 in FY2017 to \$95,181 in FY2022, a compound annual growth rate (CAGR) of 5.81% (Fig. 14) — comfortably outpacing typical U.S. CPI inflation

over the same period. The interquartile band tracks the median upward in a roughly parallel fashion, suggesting the wage growth is broad-based across the distribution rather than concentrated only at the top.

Metric	Value
Wage CAGR (FY2017 → FY2022)	5.81% / year
Starting median (FY2017)	\$71,781
Ending median (FY2022)	\$95,181

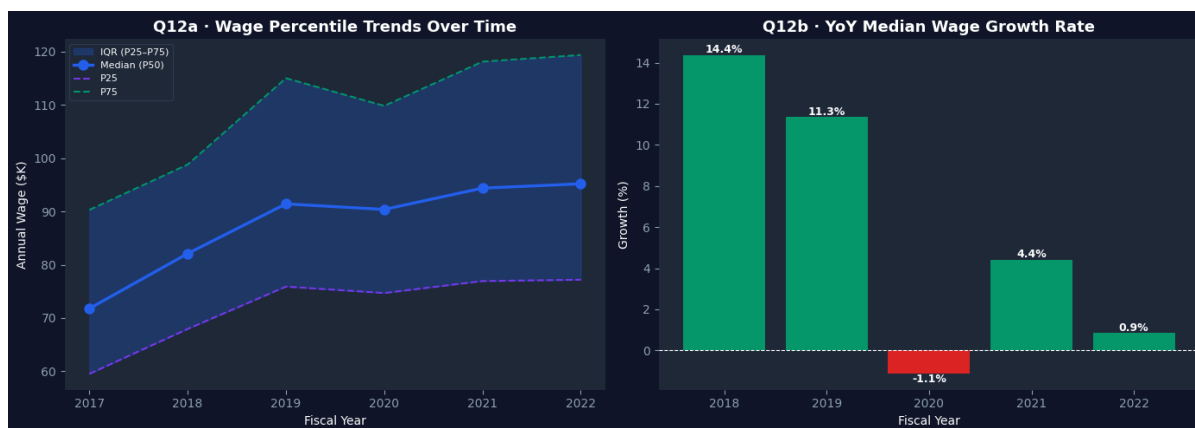


Fig. 14. Wage percentile band (P25–P75) and median trend across fiscal years.

Q13. Wage Inequality: Is the Gap Widening?

The Gini coefficient of the annual-wage distribution moved only marginally, from 0.1847 in FY2017 to 0.1841 in FY2022

(Fig. 15), indicating that within-population wage inequality among LCA filers is essentially flat — if anything, very slightly declining — over the study window, even as the

absolute wage level rose. This is a materially different conclusion from a naive reading of rising top-line wages

alone, and underscores the value of an explicit inequality metric alongside central-tendency trends.

Metric	Value
Gini FY2017	0.1847
Gini FY2022	0.1841
Trend	slightly declining (flat)

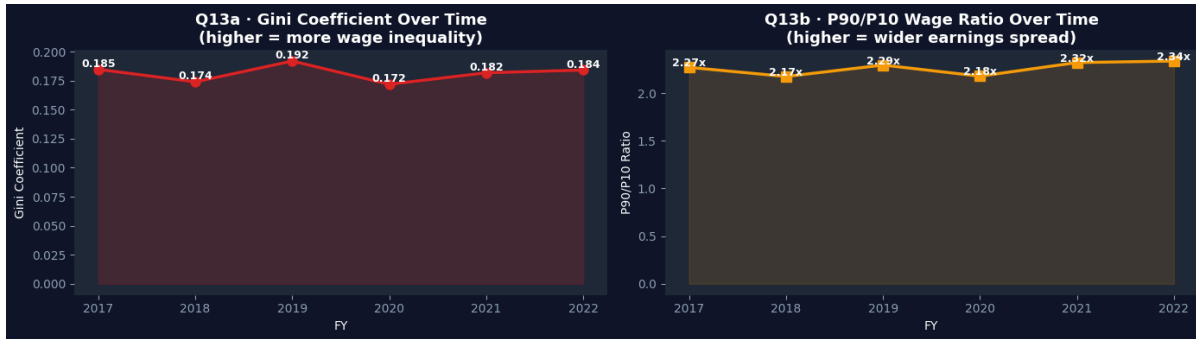


Fig. 15. Gini coefficient (left) and P90/P10 wage ratio (right) by fiscal year.

E. Theme 5 — Geographic Analysis (Q14–Q16)

Q14. Top Worksite Cities and States

Across 28,284 unique worksite locations spanning 54 states/territories, petition volume concentrates heavily in a

small number of major metropolitan labor markets — New York, New Jersey, and California lead worksite-state volume (Fig. 16), consistent with the geographic footprint of the technology, finance, and consulting sectors that are the heaviest LCA filers.

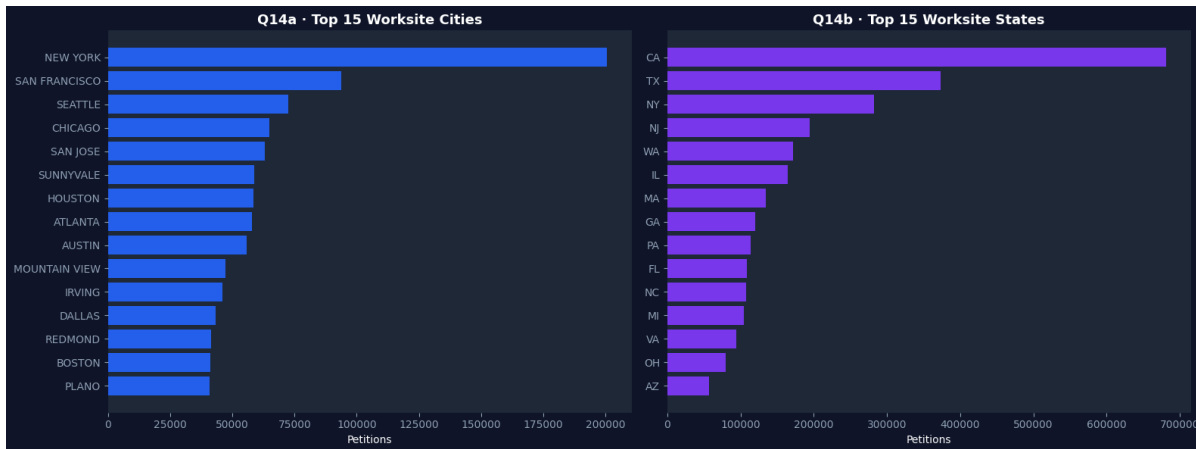


Fig. 16. Top worksite cities (left) and top worksite states (right) by petition volume.

Q15. Geographic Wage Premium (ANOVA)

Restricting to the ten highest-volume worksite states, a one-way ANOVA on annual wage by state yields $F = 37,026.15$ with $p < 0.000001$ (Fig. 17), confirming that worksite location has a statistically significant effect on

compensation. States exceeding the overall median wage (e.g., high cost-of-living technology hubs) are visually distinguished in Fig. 17 from states filing below the overall median, quantifying a clear geographic wage premium/discount structure.

Metric	Value
One-Way ANOVA F-statistic	37,026.15
p-value	< 0.000001
Result	SIGNIFICANT — location matters

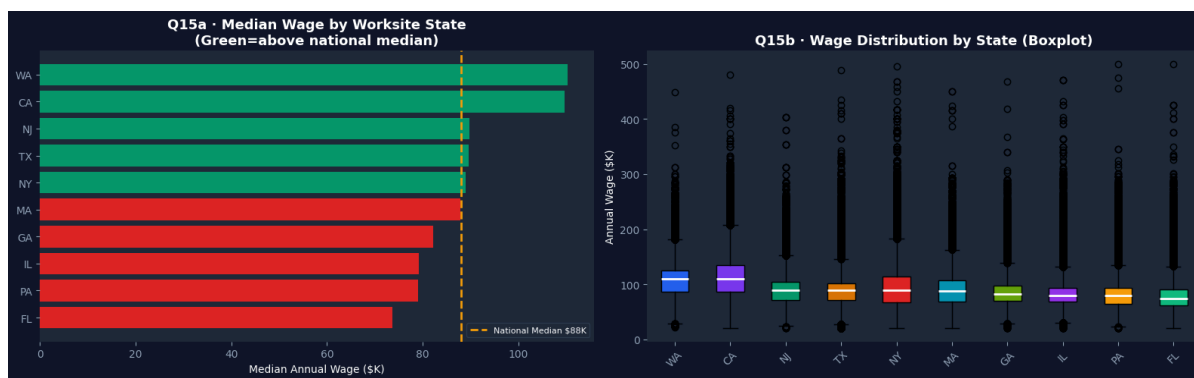


Fig. 17. Median annual wage by top-10 worksite state, with premium/discount coloring relative to the overall median.

Q16. Employer Headquarters Concentration

Ranking employer city/state pairs by petition volume (Fig. 18) reveals that employer headquarters cluster disproportionately in known outsourcing hubs such as Plano, TX and Jersey City, NJ — locations distinct from the actual worksite hotspots

(New York, San Francisco, Chicago) identified in Q14. This divergence between where petitions are filed from (employer HQ) and where the sponsored employee actually works is examined directly in Q22.

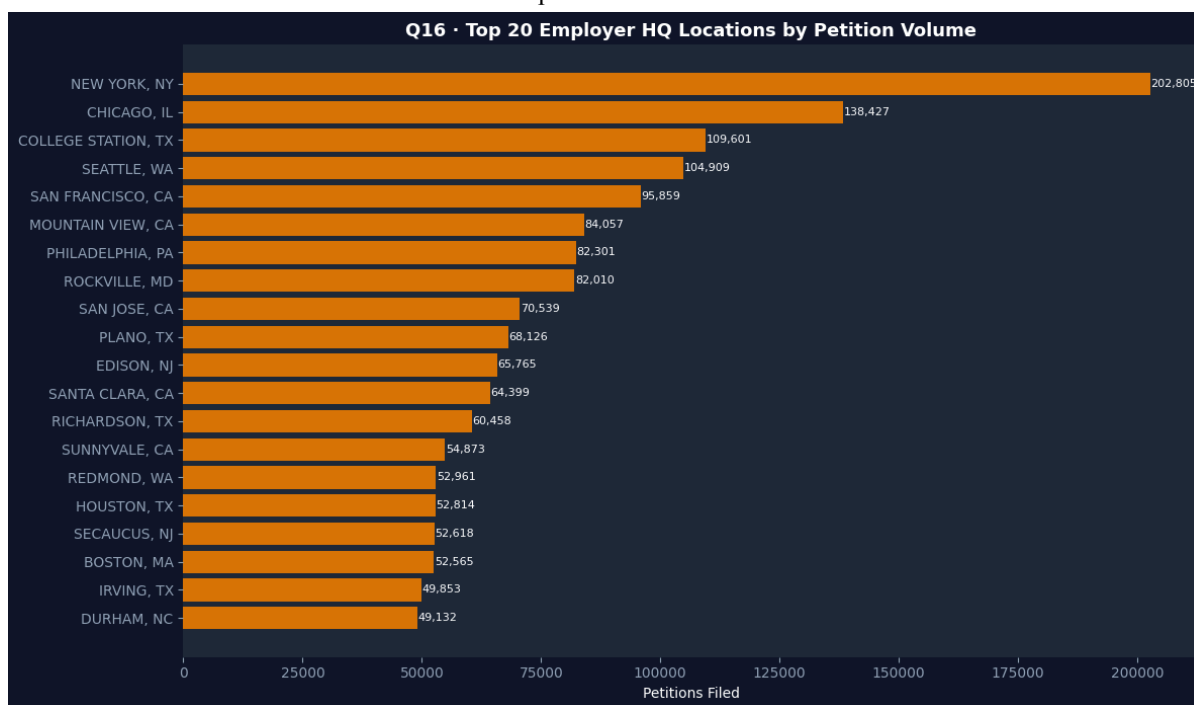


Fig. 18. Top 20 employer headquarters locations (city, state) by petition volume.

F. Theme 6 — Temporal Trends and Statistical Deep Dives (Q17, Q18, Q20)

Q17. COVID-19 Impact on Volume and Approval Rates

FY2020 (highlighted in Fig. 19) shows a visible dip in petition volume relative to the surrounding years, consistent with pandemic-driven hiring slowdowns, travel restrictions, and

processing disruptions. Notably, certification and denial rates held comparatively stable through the FY2020–FY2021 window even as volume fell, suggesting that the pandemic primarily suppressed the number of petitions filed rather than the underlying adjudication standard applied to those that were filed.

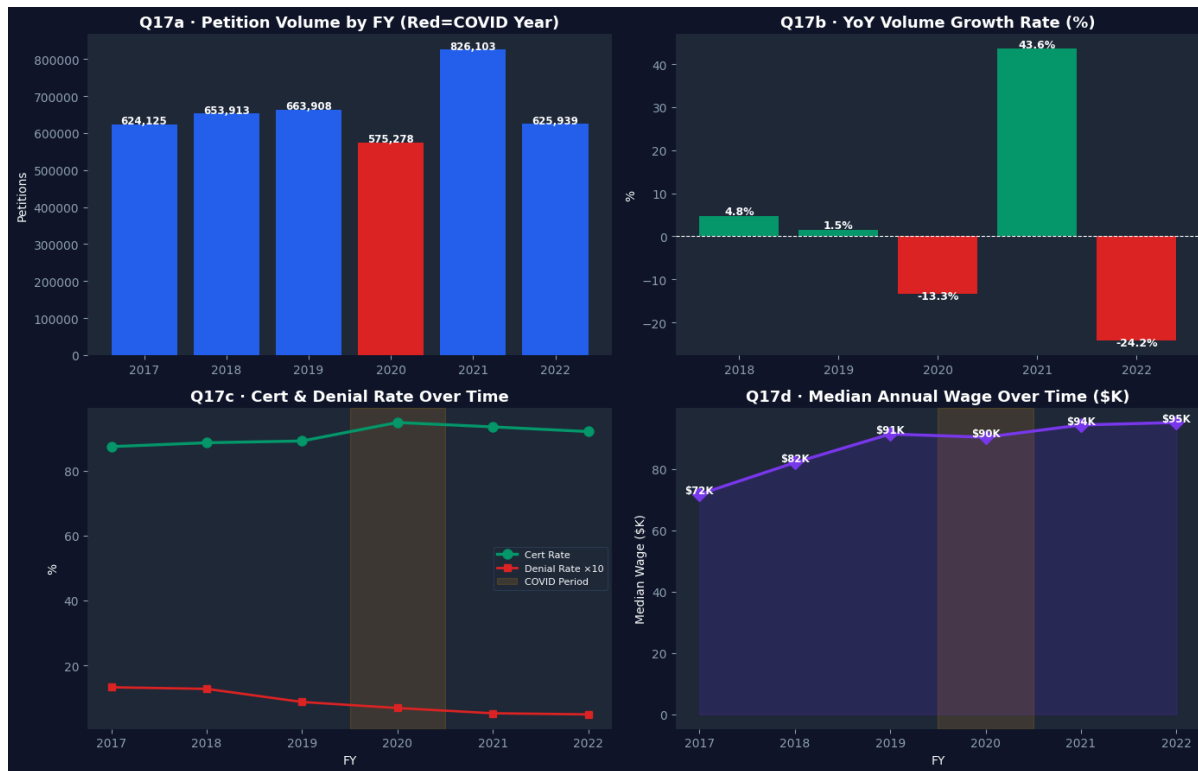


Fig. 19. COVID-era temporal dashboard: petition volume, certification rate, denial rate, and median wage by fiscal year (FY2020 highlighted).

Q18. Fastest-Growing and Fastest-Shrinking Occupations

Among the eight highest-volume SOC titles, growth from FY2017 to FY2022 is sharply bifurcated (Fig. 20): Computer Systems Engineers/Architects grew by roughly +1,269%, Software Developers (Systems Software) by +84.1%, Software Developers (Applications) by +34.8%, and Operations Research Analysts by +21.9%, while several legacy programming and analyst titles contracted sharply —

Computer Programmers fell –86.0% and “Computer Systems Analyst” (singular) fell –99.9%. The latter, near-total collapse alongside a much smaller –63.4% decline in the plural-form “Computer Systems Analysts” strongly suggests a SOC-title labeling or coding inconsistency across fiscal years rather than a genuine occupational collapse, and is flagged here as a data-quality caveat rather than a substantive labor-market finding.

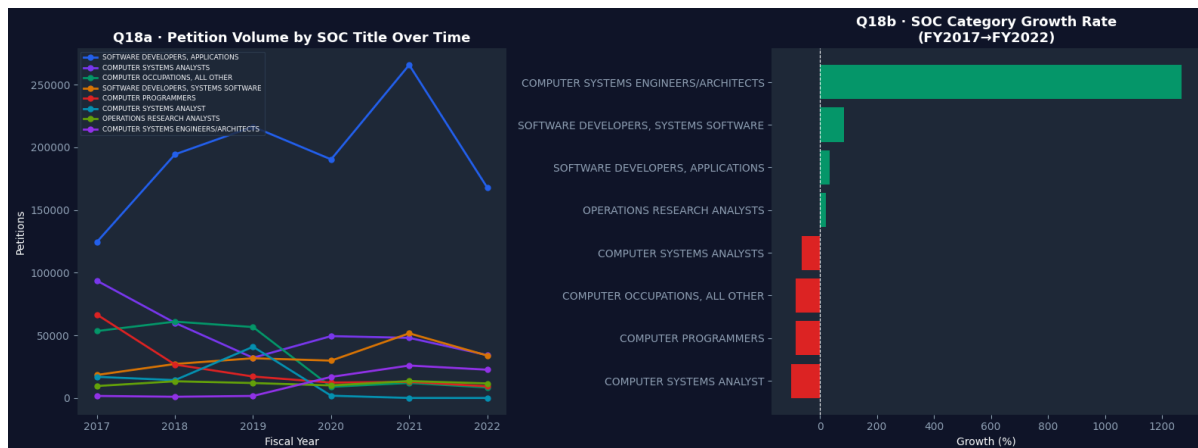


Fig. 20. SOC-title trend lines (left) and six-year growth-rate ranking (right) for the top 8 occupations by volume.

Q20. Full-Time vs. Part-Time: Wage and Volume Comparison

Full-time positions constitute the overwhelming majority of filings (≈98%), yet even after annualizing wages, full-time

median pay (\$88K) exceeds part-time median pay (\$73K) by a wide and statistically significant margin (Welch's $t = 83.759$, $p < 0.000001$; Fig. 21), indicating that the full-time/part-time wage gap is not merely an artifact of pro-rating hours.

Metric	Value
Welch's t-test (Full-Time vs Part-Time)	
Full-Time median	\$88K
Part-Time median	\$73K
t-statistic	83.759
p-value	< 0.000001
Result	SIGNIFICANT

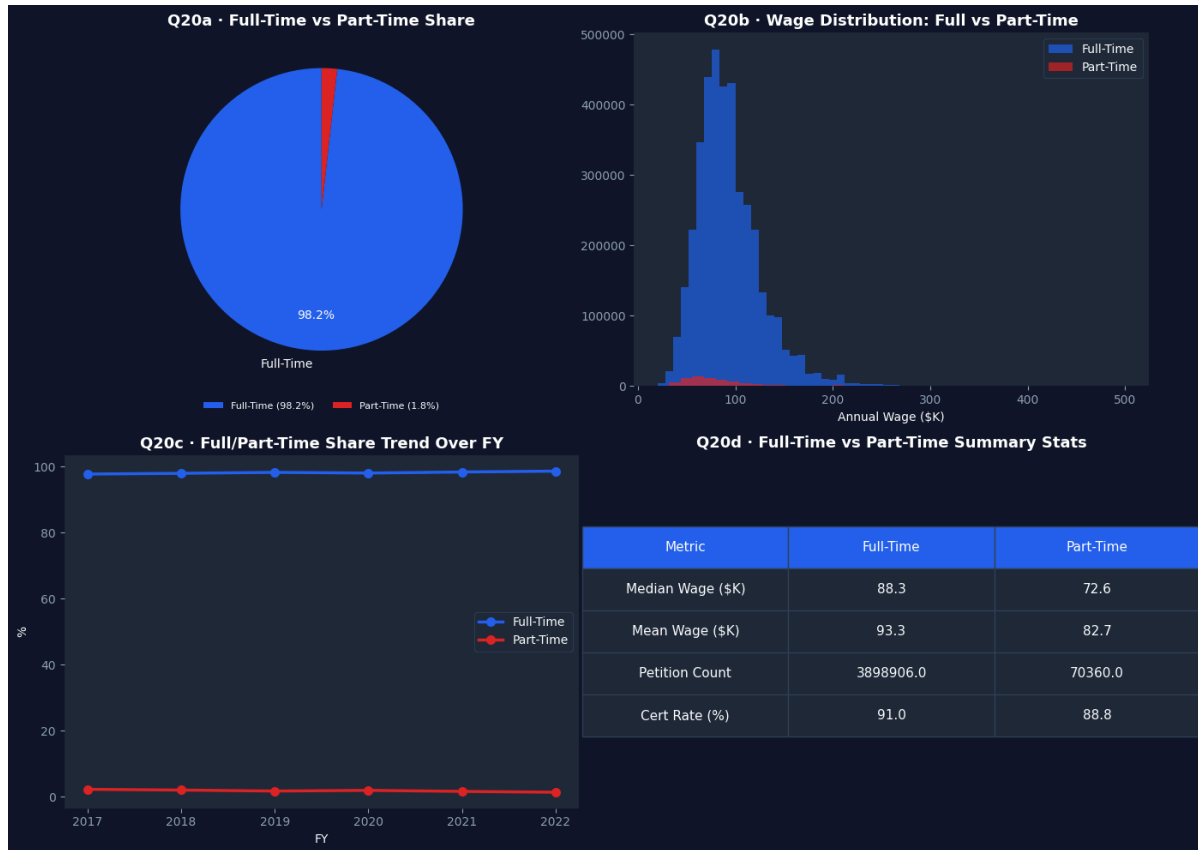


Fig. 21. Full-time vs. part-time comparison: filing share, wage distributions, and fiscal-year trend.

G. Theme 7 — Employer Location Deep Dive (Q21–Q24)

Q21. Employer Headquarters Concentration (HHI)

Using the corrected EMPLOYER_STATE field, the Herfindahl-Hirschman Index (HHI) of employer-HQ concentration across states is 860 (Fig. 23), well below the 1,500 threshold conventionally used to denote a

competitive/dispersed market and far from the 10,000 value denoting a single-state monopoly. Employer headquarters are therefore geographically dispersed at the state level, even though Q16 shows sharp concentration at the city level within a handful of dominant metro hubs.

Metric	Value
Employer-state HHI	860
Interpretation: Competitive / dispersed employer-HQ landscape	

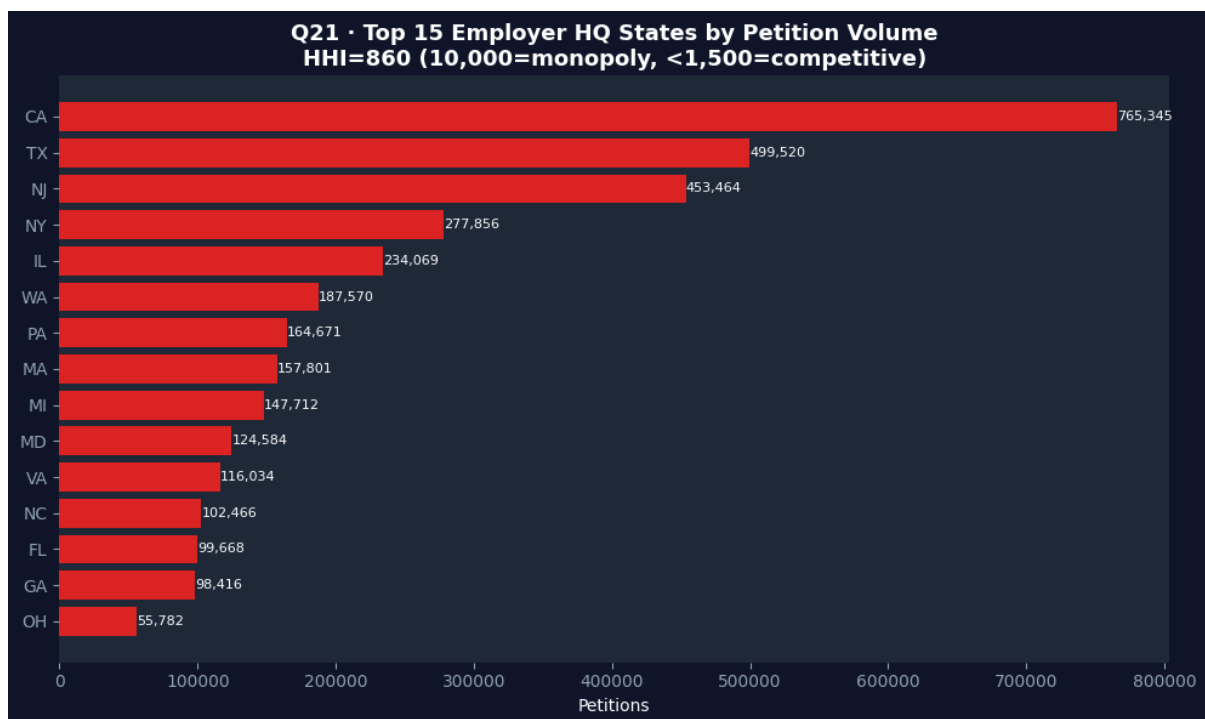


Fig. 22. Top 15 employer headquarters states by petition volume, annotated with the computed HHI.

Q22. Worksite/HQ State Mismatch (Remote and Staffing Proxy)

Among petitions with both fields populated, 48.3% place the sponsored worker in a state different from the employer's headquarters state (Fig. 24) — a strong proxy signal for consulting- or staffing-style placement models (in which an

IT-services firm headquartered in one state places workers on client sites nationwide) as opposed to co-located, in-house hiring. This nearly-even split is consistent with the employer-HQ vs. worksite divergence already observed qualitatively in Q16.

Metric	Value
Worksite state $\neq$ Employer HQ state	48.3% of petitions

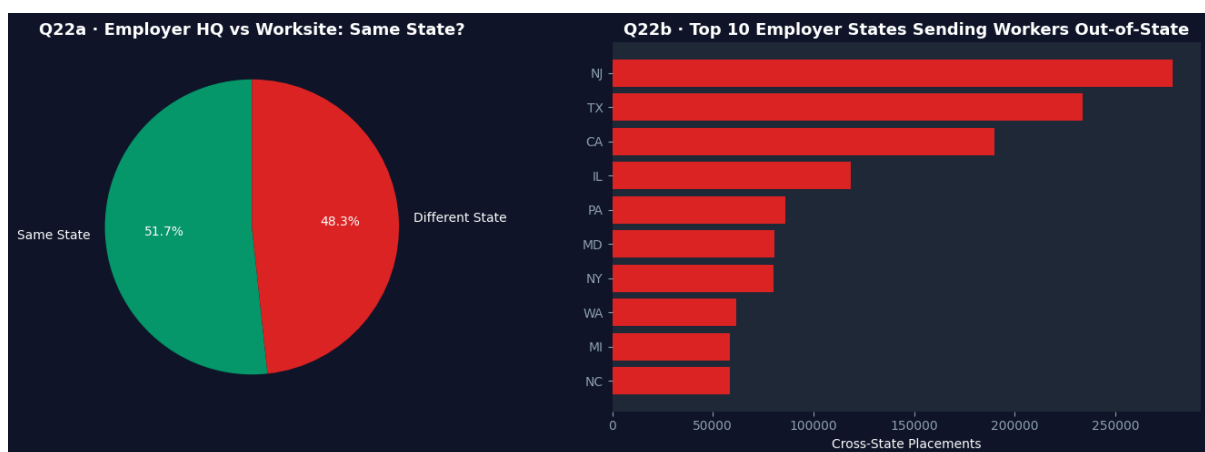


Fig. 23. Share of petitions with worksite state matching vs. differing from employer headquarters state.

Q23. Wage Volatility by Occupation (Coefficient of Variation)

Ranking SOC titles with at least 200 petitions by coefficient of variation (CV = std/mean) reveals that certain occupations

exhibit far wider relative pay dispersion than others (Fig. 25). High-CV roles typically span a broad range of seniority levels and geographic markets under a single occupational code, while low-CV roles tend to be more standardized,

entry-consistent job functions with narrower internal pay bands.

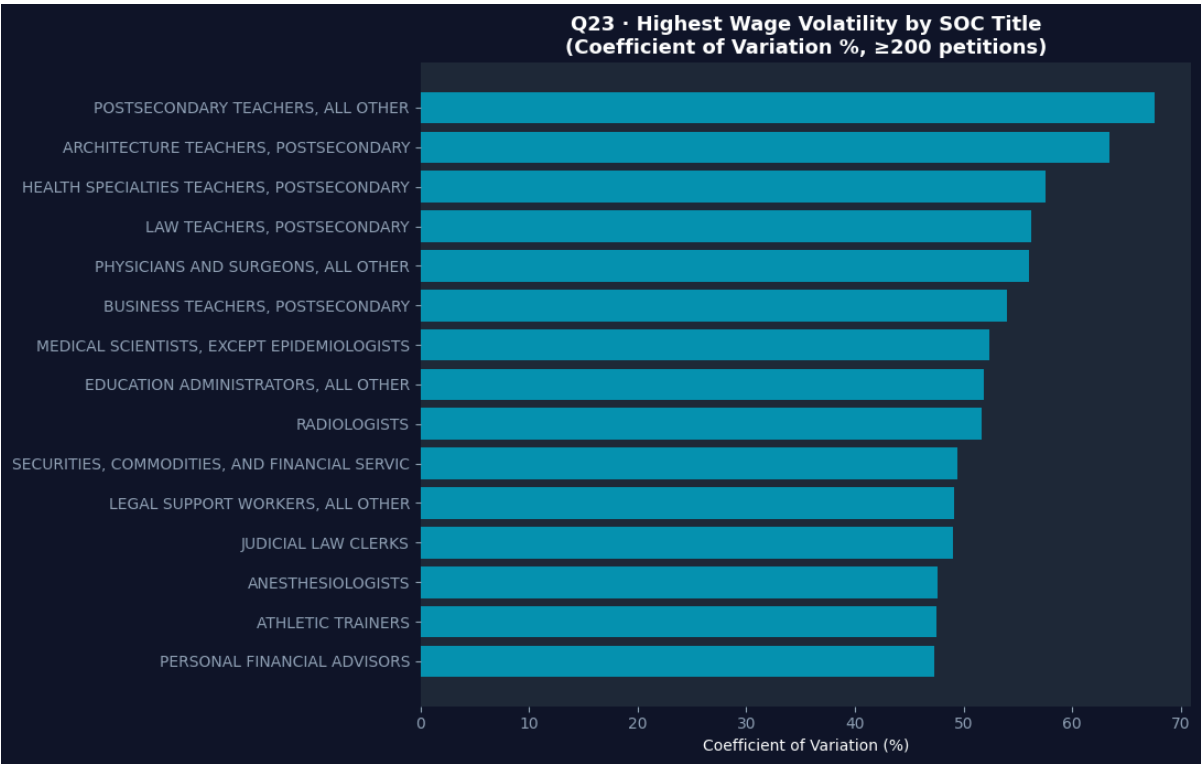


Fig. 24. Top 15 SOC titles by wage coefficient of variation (≥200 petitions).

Q24. H-1B vs. E-3 Wage Gap Within the Same Occupation

A direct same-occupation comparison of H-1B and E-3 median wages was attempted by restricting to SOC titles with at least 100 petitions in each visa class. In the current extract, an insufficient number of SOC titles met this joint sample-size threshold to support a statistically reliable comparison, and no figure is reported for this question. This null result is reported transparently rather than omitted, and is best interpreted as reflecting the much smaller absolute volume of E-3 filings relative to H-1B (consistent with the visa-class composition established in Q4) rather than as evidence of an absent wage gap.

IV. LIMITATIONS

Several limitations qualify the findings above. First, LCA data reflect employer attestations at filing time and do not confirm that a petition was ultimately approved by USCIS, that the worker was admitted, or that the offered wage was actually paid. Second, several free-text fields (JOB_TITLE, EMPLOYER_NAME) exhibit extreme cardinality and likely contain unresolved duplicate entities under differing spellings, which was only partially mitigated by relying on the standardized SOC_TITLE field. Third, the SOC-title labeling inconsistency identified in Q18 (singular vs. plural “Computer Systems Analyst[s]”) suggests that occupation-level growth-rate figures should be interpreted cautiously wherever near-total collapses or explosive growth rates are observed

without corroborating evidence. Finally, the Q24 same-occupation H-1B/E-3 wage comparison could not be completed at adequate sample size, and any future extension of this work would benefit from a longer time window or a relaxed minimum-count threshold to enable that comparison.

V. CONCLUSION

This report has presented a rigorous, twenty-three-question empirical case study of nearly four million U.S. Labor Condition Application records spanning FY2017–FY2022. The evidence assembled here supports several robust, statistically validated conclusions: H-1B overwhelmingly dominates the specialty-occupation visa landscape; a small number of large IT-outsourcing employers file a disproportionate and systematically lower-paid share of petitions; annual wages carry a statistically significant geographic premium and have grown at a real CAGR of 5.81% per year without a corresponding rise in within-population wage inequality; and nearly half of all petitions place workers outside their sponsoring employer's home state, consistent with a substantial staffing/consulting placement model. Collectively, these findings offer a quantitatively grounded foundation for further research and policy analysis on the economics of U.S. skilled-worker immigration.